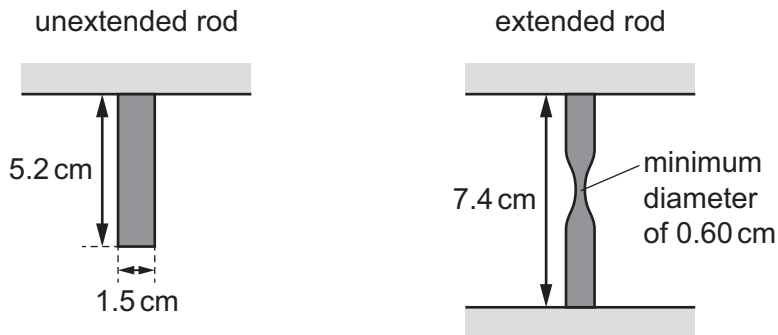


20 A cylindrical steel rod with negligible weight has a diameter of 1.5 cm and a length of 5.2 cm.

The rod is firmly attached at the top end.

The rod is firmly attached to a machine at the other end that applies a constant force of 363 N to the rod.

This causes the rod to extend to a length of 7.4 cm and to have a minimum diameter of 0.60 cm in the position shown.



What is the maximum stress acting on the steel rod when the length is 7.4 cm?

- A** $2.1 \times 10^6 \text{ Pa}$ **B** $3.2 \times 10^6 \text{ Pa}$ **C** $5.7 \times 10^6 \text{ Pa}$ **D** $1.3 \times 10^7 \text{ Pa}$